

Same Product, Different Visibility: Representation Robustness in Neural E-Commerce Retrieval

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Abstract

Neural retrieval increasingly mediates which Web content can be discovered, compared, or used by agentic interfaces. Semantically equivalent information should ideally receive comparable retrieval opportunities, yet structured records are often flattened into ordered text. We study this representation robustness problem in product search, holding facts and relevance fixed while permuting attributes. Across two benchmarks and three general-purpose retrievers, 4.4–13.3% of highest-relevance pairs cross Top-20 under catalog-wide permutations. Target-only interventions confirm that the same product can change visibility even with all competitors fixed. We then ask whether fact-preserving strategies improve relevant-product recall, comparing fixed ordering, full-record aggregation, set encoding, and lexical/hybrid retrieval under equal unique-candidate budgets. A development-selected ordering rule does not yield consistent recall improvements and loses recall when transferred to another encoder. Set encoding eliminates measured order instability, while its condensed-NDCG changes are small and statistically inconclusive; this does not establish equivalence or preserved recall. Candidate-budget and fixed-text reranking analyses separate useful retrieval from stable omission: a reranker cannot recover content it never receives. These results support evaluating representation robustness jointly with relevant-content recall, transfer, and computational cost.

CCS Concepts

• **Information systems** → **Information retrieval**; *Recommender systems*.

Keywords

Web retrieval, product search, robustness, representation invariance, visibility

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1 Introduction

Web users increasingly reach information through systems that decide what to retrieve before the user sees it. Search, recommendation, retrieval-augmented systems, and emerging agentic interfaces all depend on an initial selection of content. For a shopping agent to compare a relevant product, that product must first enter its candidate set. This motivates a retrieval question, without assuming that better retrieval necessarily improves recommendations or purchases: *can irrelevant choices in how unchanged information is represented determine whether relevant content is found?*

Much of the Web is structured or semi-structured. Products, jobs, hotels, and events are stored as fields, yet neural retrieval often consumes flattened token sequences. Many underlying facts have

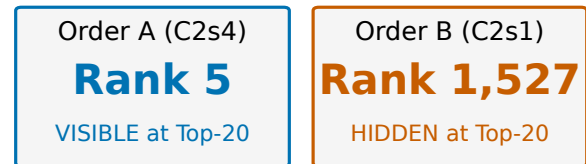
no meaningful linear order. Product color, material, dimensions, and compatibility remain the same facts when their positions in a record change. This mismatch makes serialization a potential nuisance signal between semantic relevance and retrieval visibility.

E-commerce provides a controlled setting for studying that mismatch. We preserve product identity, all source facts, the query, the model, and the relevance labels while permuting complete attribute atoms. We distinguish a catalog-wide intervention, corresponding to a shared indexing change, from a target-only intervention that holds every competitor in its original representation. The latter supports the precise comparison: same product, same facts, same competitors, same query, same model; only the target serialization changes.

Figure 1 illustrates the effect. For *turquoise chair*, an Exact-relevant product moves between ranks 5 and 1,527 under different attribute orders with every competitor fixed. Its original rank is 486. The example is selected for interpretability and a large crossing; aggregate results, rather than this extreme case, establish the population pattern. The full title, source atoms, identifiers, and all tested ranks are included in the artifact and appendix.

Query: "turquoise chair"

Same Exact-relevant product and facts
Same model • Every competitor fixed in C0



Only the target attribute order changes
Should serialization determine visibility?

Figure 1: Target-only intervention, WANDS query 162 and product 34536, MiniLM. The same Exact-relevant product crosses Top-20 with competitors fixed in C0. This differs from catalog-wide reserialization.

Order instability is important, but eliminating it is not sufficient. A relevant product that is never retrieved under any representation has zero crossing instability. We therefore deepen the robustness question into a recall-first evaluation: *can fixed, fact-preserving representation strategies make relevant products more reliably retrievable on queries not used for strategy selection, and at what cost?* Highest-label Recall@100 is the primary strategy outcome, with relevance effectiveness, instability, and resource costs reported alongside it.

The available intervention also matters. A merchant can change one product’s text, whereas a platform can change an entire index, aggregate embeddings, or store multiple vectors per product. A target-only gain does not establish a market-wide benefit, and a platform-only method is not a merchant-deployable rewriting rule. We compare these settings explicitly and prevent multi-view retrieval from receiving extra candidate slots: all methods return the same number of unique products.

Our contribution is an empirical diagnosis and controlled strategy evaluation, not a claim to invent centroid averaging, multi-vector retrieval, or rank fusion. We ask four questions: how equivalent representations affect relevant-product inclusion; where instability concentrates after accounting for rank and input length; whether development-selected strategies transfer; and how their recall and cost compare with simply enlarging the candidate pool. We combine cross-model sensitivity, single-product interventions, rank-conditioned diagnostics, strong lexical/hybrid controls, and fixed-text reranking. Negative strategy results and the distinction between stability and useful retrieval remain central to the interpretation.

2 Related Work

2.1 Structured Product and Hybrid Retrieval

Product search must match queries to heterogeneous catalog fields. Lexical retrieval such as BM25 provides a strong order-insensitive baseline under a fixed token multiset [15]; dense matching and field-aware models address semantic mismatch [6]. Attribute extraction can supply explicit matching signals [12]. CHARM constructs hierarchical field representations with block-triangular attention and uses aggregated vectors for candidate retrieval followed by richer field-level matching [8]. Its learned, field-aware hierarchy differs from our frozen-encoder views of complete, fact-equivalent product records. We use it to position multi-vector product retrieval, not as an implemented baseline.

Hybrid retrieval combines lexical and semantic signals. Reciprocal rank fusion combines rankings without requiring comparable score scales [7]. We include fixed full-list RRF as a practical competitor because a representation strategy is less compelling if a simple hybrid already obtains better recall. Our comparisons hold candidate count fixed and separately report the extra storage and scoring cost of multiple product vectors. The purpose is to evaluate the representation decision, rather than propose another fusion algorithm.

2.2 Content and Visibility Optimization

GEO studies how content changes influence visibility in generative systems [2]; E-GEO specializes this objective to e-commerce [3], while SAGEO Arena evaluates search-augmented optimization in a broader retrieval-and-generation setting [11]. Ranking-incentivized content modification likewise studies strategic changes under quality constraints [9]. Our initial question is different: why should a semantically irrelevant representation choice alter visibility in the first place? Our subsequent strategy evaluation preserves raw facts and uses query-independent rules or platform indexing. We do not evaluate generated product claims, agent recommendations, purchases, or merchant welfare.

2.3 Representation Robustness

Serialization sensitivity is not a new discovery in neural retrieval. Bhandari et al. study alternative table serializations, centroid representations, and a residual adapter toward centroid targets [4]. Their centroids motivate our full-record averaging baseline, which is distinct from independent attribute encoding followed by set pooling. We extend the empirical question to judged-relevant product inclusion, target-only causal control, persistent omission, cross-encoder candidate limitations, and recall–cost tradeoffs. We do not claim that a finite mean over views is a new invariant-learning algorithm.

Order effects and behavioral retrieval testing also caution against treating aggregate effectiveness as a complete description of a neural system [1, 13]. Our task-defined equivalence class is deliberately narrower than arbitrary rewriting: complete raw attribute atoms are moved while their contents and multiplicities remain fixed. Generalization to other structured Web domains remains an empirical question.

3 Problem Formulation

Let $F(p)$ denote the information underlying item p , and let $\mathcal{R}(F(p))$ contain representations that preserve that information for the task. Human relevance $\text{Rel}(q, p)$ remains fixed across the class. A normalized dual encoder scores a representation by $z(q, r(p)) = \langle E_q(q), E_p(r(p)) \rangle$. A representation-robust retriever should ideally avoid substantial visibility differences induced solely by task-irrelevant changes in r .

For rank $\rho_r(q, p)$ and cutoff K , define

$$\text{VI@}K(q, p) = \mathbb{I} \left[\min_{r \in \mathcal{R}} \rho_r(q, p) \leq K < \max_{r \in \mathcal{R}} \rho_r(q, p) \right]. \quad (1)$$

VI is an any-in/any-out event over the tested family, not a single-update loss probability. We also partition relevant pairs into always retrieved, sometimes retrieved, and never retrieved. Sometimes retrieved equals crossing; never retrieved can be a serious recall failure despite VI=0. Rank range describes movement independently of a cutoff. VI need not increase with K , since a boundary can move beyond an item’s entire rank interval.

Let H_q be the judged highest-relevance products for query q . Our primary strategy metric is

$$\text{Recall@}K(q) = \frac{|H_q \cap \text{Top}K(q)|}{|H_q|}. \quad (2)$$

Queries with $|H_q| = 0$ have undefined recall and are excluded only from recall aggregation. Query-macro averages weight eligible queries equally; pair-micro statistics weight relevant pairs equally. For each method–Original comparison we retain rescued and newly missed relevant pairs; their difference, normalized by $|H_q|$, is the per-query recall change.

Relevance–visibility alignment requires useful retrieval as well as stability. We report condensed NDCG, which removes unjudged products before discounting, separately from complete-catalog recall and rank visibility [10, 16]. Unknown labels are not treated as irrelevant. In a two-stage pipeline, representation-dependent Irrecoverable@100 counts products present in some first-stage pools but absent in others; persistent omission is reported separately. Ranker VI on common candidates is conditional on survival and cannot be compared directly with all-highest-pair VI.

4 Interventions and Retrieval Strategies

4.1 Equivalent Inputs and Intervention Permissions

The primary family contains C0 source order, C1 reversal, and five deterministic attribute permutations. WANDS atoms are raw key-value fields; ESCI uses available brand, color, and complete bullet-point strings as atoms, retaining the internal bullet text. Audits preserve complete raw atoms and multiplicity, non-attribute fields, source values, and character/lexical token multisets. Catalog-wide intervention changes all products. Target-only intervention changes one relevant target at a time with every competitor in C0:

$$\rho_s^{\text{target}}(q, p) = 1 + \sum_{u \neq p} \mathbb{I}[z(q, C0(u)) > z(q, s(p))], \quad (3)$$

with stable catalog-order tie handling. The old target entry is removed before insertion. These interventions answer different questions; their crossing rates do not form an additive causal decomposition.

Merchant-side strategies must produce one complete product text without knowledge of a future query. We freeze four field-priority rules emphasizing type/style, appearance, use/compatibility, or dimensions/material. They move entire raw atoms, retain the original free text and section order, and use a canonical lexical fallback. Matching uses field keys where available and literal prefixes for unkeyed ESCI atoms. One global rule is chosen on development Recall@100; cNDCG and a fixed simplicity order break ties. No product-specific or category-specific supervised policy is fitted.

We distinguish pure raw-atom sorting from the historical canonical template M2, which also adds field framing and changes section placement. Both are retained as baselines. This avoids attributing the template’s effect entirely to sorting. The selected priority rule is evaluated both catalog-wide and target-only; the latter is a diagnostic of a single record’s effect, not a deployment policy that knows test relevance labels.

4.2 Platform-Side Aggregation and Fusion

For full-record multi-view methods, generate nested sets of $m = 2, 4, 7$ views by shuffling a canonical raw-atom list using fixed product-ID-dependent seeds. Each complete record is encoded and normalized independently. Centroid retrieval uses

$$v_m(p) = \text{norm} \left(\frac{1}{m} \sum_{j=1}^m E_p(r_j(p)) \right), \quad (4)$$

whereas multi-vector retrieval scores p by $\max_j \langle E_q(q), E_p(r_j(p)) \rangle$. Reduction is by product before selecting K unique products. Equal m is used for every product; actual distinct text and encoded-view counts are recorded. The canonical-anchored generator is independent of incoming attribute order. This does not make finite averaging an exhaustive orbit average or establish robustness to other semantic transformations.

The existing set-mean method instead encodes individual attributes. With normalized non-attribute vector $t(p)$, it computes

$$a(p) = \text{norm} \left(\frac{1}{|A(p)|} \sum_{a \in A(p)} E_p(a) \right), \quad (5)$$

$$v(p) = \text{norm}(.5t(p) + .5a(p)).$$

The attribute mean is normalized before interpolation, matching the implementation; the earlier draft omitted this intermediate step. Empty attribute sets contribute zero. This proof of concept removes order dependence but may lose useful cross-attribute interactions. It is evaluated for recall and effectiveness, not declared successful because VI is zero.

BM25 uses the full lowercase alphanumeric token multiset without positional features or truncation. Hybrid retrieval adds full-list reciprocal-rank scores with constant 60 [7]. These are platform controls. Multi-vector and centroid methods increase offline encoding; only the former retains multiple index vectors. None is represented as a merchant-editable text strategy.

5 Experimental Setup and Audit

5.1 Data, Splits, and Evaluation History

The historical WANDS catalog contains 42,994 products, 480 queries, and 231,859 deduplicated, noncontradictory judgments. Highest relevance is Exact, with declared gains 3/1/0 for Exact/Partial/Irrelevant. The historical ESCI experiment uses 500 US small-version test queries and a fixed union of 10,076 products with 10,134 judgments [5, 14]. We report judged-relevant recall; incomplete judgment coverage prevents claims about every truly relevant product.

The extension freezes the existing WANDS development split: 96 queries selected by SHA256 order and 384 held-out queries. Historical test results have already been inspected, so these are post-hoc fixed-strategy evaluations, not pristine confirmation. Only development labels choose the global rule and the best added configuration among the rules, centroid2/4, and max2/4. The chosen configuration transfers unchanged to MiniLM and ESCI; $m=7$ is a saturation check and cannot replace the winner. All screening results, including losses, remain available.

A further 100 ESCI query IDs absent from the historical query set are frozen before retrieval. Their judged products are added to the historical union to define a new common catalog for all compared methods. This evaluates newly sampled queries, while acknowledging prior access to the source dataset. Absolute recall differences from the old catalog cannot be attributed to generalization alone because competition changes. No product-disjoint or unseen-category claim is made. The new catalog contains 12,019 products and 1,974 judgments. Recall/cNDCG eligible query counts are 308/383 for held-out WANDS, 499/500 for historical ESCI, and 100/100 for the new queries.

5.2 Models and Reconciliation

We retain pinned MiniLM, BGE, GTE, and the historical product-specific encoder for sensitivity evidence; exact model identifiers and revisions are in the artifact. New strategy selection uses frozen BGE; MiniLM tests transfer to another encoder family without retuning.

BGE uses CLS and 512 tokens, MiniLM mean pooling and 256 tokens. Query prefixes remain empty, matching the historical condition; BGE v1.5 supports instruction-free use although its model card recommends an instruction for retrieval. CUDA weights are FP16 with FP32 pooling, normalized vectors, and exact stable catalog-order ranking. Models are not retrained.

The audit found that historical ESCI gains interchanged Substitute and Complement. The benchmark paper specifies gains 1, .1, .01, and 0 for E, S, C, and I [14]. Its training code agrees, but the pinned repository’s qrels/evaluation-shell combination is internally inconsistent with that definition. We therefore recompute effectiveness from unchanged ranks under the paper’s gains and retain the historical values in a reconciliation artifact. Highest-label recall, ranks, and VI do not depend on that swap. The earlier mitigation join repair is preserved: cNDCG retains every positive-IDCG query, independently of highest-label availability.

5.3 Uncertainty, Cost, and Reranking

Primary strategy comparisons use query-macro Recall@100 and paired query bootstrap with 10,000 resamples. We report fixed contrasts against Original and hybrid, the selected rule against Original, and direct set-mean–canonical comparisons. Confidence intervals spanning zero do not establish equivalence; no non-inferiority margin is claimed. Diagnostic rank strata use common-query support and clustered query uncertainty, rather than treating all pairs as independent.

We reuse full rankings for $K = 20, 50, 100, 200, 500, 1000$. Local measurements use an Intel i7-13700H CPU and RTX 4060 Laptop GPU with 8 GB memory. Cost separates dense-vector/sparse-posting storage, query encoding, and exact scoring/sorting on four CPU threads. Repeated timing uses the first 50 queries, a warm-up, and five measured repetitions. Encoding costs and distinct-view counts are retained in the artifact. These are local measurements, not ANN or commercial-scale latency estimates.

The new reranker experiment uses the pinned MiniLM cross-encoder with one fixed M2 canonical text per product, joint 256-token input, $K = 50, 100, 200$, and a final Top-20. Candidate-union caching avoids repeated inference. Returned-list condensed NDCG removes unjudged items from those final results but normalizes against all judged ideal gains; it is distinguished from full-catalog cNDCG. The historical experiment, which also varied reranker input, is retained as a separate condition. Its highest-relevance Top-100 memberships survive reconciliation even where floating-point and tie handling alter other candidate positions.

6 Results

6.1 RQ1: How Does Representation Affect Relevant-Product Inclusion?

Fact-equivalent orders repeatedly move judged-relevant products across retrieval cutoffs. Across the three general encoders and two datasets, catalog-wide pair-micro VI@20 ranges from 4.42% to 13.26%. Figure 2 presents query-macro estimates and uncertainty, rather than duplicating every statistic in a main-text table. The e-commerce-specific encoder also remains sensitive: its pair-micro

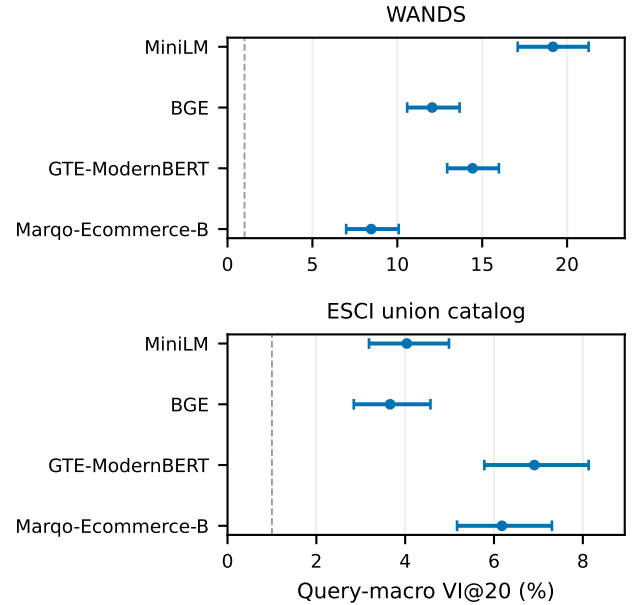


Figure 2: Catalog-wide query-macro VI@20 with query-bootstrap 95% CIs. General and product-specific retrievers remain sensitive. The dashed 1% reference is a historical study threshold, not a preregistration of this extension.

VI@20 is 6.07% on WANDS and 7.22% on ESCI. These are seven-representation crossing rates, not expected losses from one arbitrary update.

The primary-family target-only intervention fixes every competitor in C0. WANDS pair-micro VI@20 is 12.72%, 8.84%, and 11.39% for MiniLM, BGE, and GTE; ESCI values are 3.11%, 3.16%, and 6.68%. The difference from catalog-wide results reflects the intervention, not a contradiction. Figure 1 uses this stricter condition. Its ranks 5 and 1,527 must not be confused with the old catalog-wide ranks 3 and 1,303. A merchant changing one record and a platform changing all records face different competition.

Instability omits persistent failure. Across all historical queries at Top-100, BGE retrieves 8,384 WANDS highest-label pairs under every order and 3,856 under only some orders; 13,230 never enter any tested pool. ESCI has 3,995 always retrieved, 44 sometimes retrieved, and 395 never retrieved pairs. The sometimes-retrieved population is representation-dependent candidate loss; the never-retrieved population has VI=0 despite complete failure at this budget. This is why the strategy evaluation below prioritizes recall. Appendix tables retain all cutoffs and denominators.

6.2 RQ2: Where Does Instability Concentrate?

Distance to the retrieval boundary is a strong interpretive constraint. An irrelevant product far below Top-20 has little opportunity to cross, so an unadjusted relevance-stratum difference cannot establish preferential harm to relevant products. Figure 3 compares highest and irrelevant labels within the same original-rank band and query, reporting only their common support.

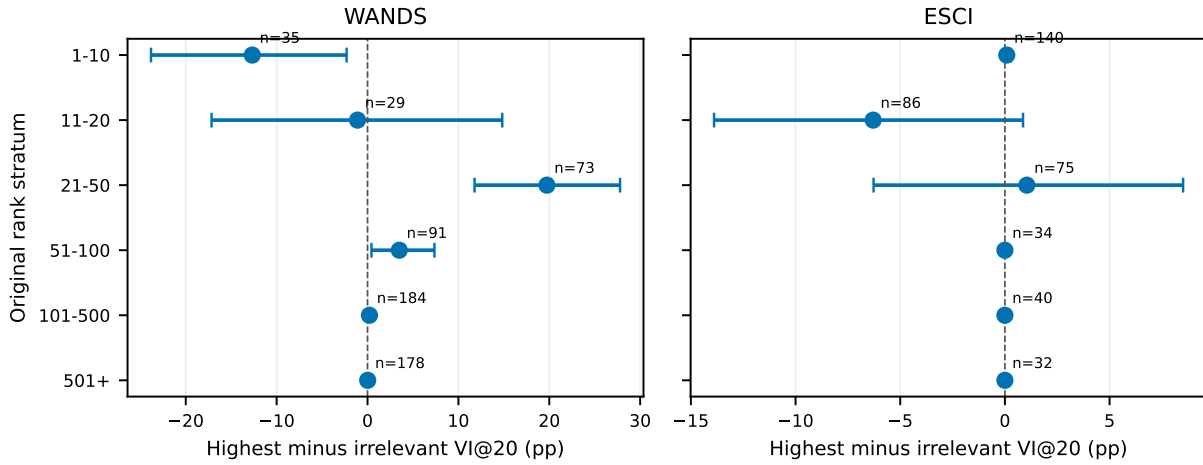


Figure 3: Rank-conditioned highest-minus-irrelevant query-macro VI@20, in percentage points. Intervals use query bootstrap; n is common-query support in each band. These coarse strata do not eliminate residual score-margin confounding.

On WANDS the contrast is negative for original ranks 1–10 (−12.68 points, CI [−23.83, −2.30]) but positive for ranks 21–50 (+19.74, CI [11.78, 27.79]). ESCI does not show a consistent positive contrast across strata. We therefore weaken the earlier unadjusted interpretation: relevant products clearly suffer instability, but the data do not establish a universal relevance-specific gradient after accounting for rank. These are descriptive associations, not fairness or disparate-impact results.

Truncation does not explain every crossing. Among WANDS Exact pairs whose complete inputs fit BGE’s token budget in all seven orders, query-macro VI@20 is 8.69% (CI [7.15, 10.39]; 8,536 pairs, 292 queries). The over-budget subset is 12.77% ([10.90, 14.75]; 16,934 pairs, 343 queries). The two populations differ in more than truncation, and queries may appear in both. Their contrast is not a causal truncation effect; the fitting subset establishes sensitivity even with complete input coverage.

The artifact also stratifies by score margin, attribute count, maximum token length, duplicate atoms, query length, and literal query–attribute-value overlap. Lexical overlap is an auditable surface feature, not a validated query-intent taxonomy. Historical chunking and pooling controls likewise retain nonzero instability. Together these checks narrow explanations without isolating attention, position, or truncation as a unique mechanism.

6.3 RQ3: Do Fixed Strategies Improve Relevant-Product Recall?

Development selects appearance-first from the eight frozen added configurations, with Recall@100 0.680; Original is 0.662 and hybrid 0.717. The independently selected global rule is appearance-first. Being the best added configuration does not imply beating either baseline. The full screening table is retained, and the choice is frozen before transfer evaluation.

On held-out WANDS, the selected strategy changes Recall@100 by −0.05 [−1.21, +1.12] percentage points relative to Original and

Table 1: Fixed BGE strategies. R100 is query-macro highest-label recall; cN is cNDCG@10. W/E denote held-out WANDS and the historical ESCI union. ESCI uses benchmark-standard gains.

Method	R100 W	R100 E	cN W	cN E
Original	0.632	0.910	0.825	0.736
BM25	0.632	0.828	0.787	0.706
BM25 + dense	0.681	0.915	0.827	0.727
Raw sort	0.632	0.910	0.823	0.738
Canonical template	0.633	0.913	0.803	0.742
Set mean	0.618	0.890	0.815	0.733
Appearance-first	0.631	0.911	0.823	0.738

Table 2: Unchanged BGE-selected strategy transferred across evaluation cells. Entries are Recall@100 differences from Original in percentage points, with paired query-bootstrap 95% CIs. Historical evaluations were previously inspected; new ESCI queries use a separate common catalog.

Data	Encoder	Δ	95% CI
WANDS	BGE	−0.05	[−1.21, +1.12]
ESCI old	BGE	+0.08	[−0.04, +0.19]
WANDS	MiniLM	−1.75	[−3.00, −0.63]
ESCI old	MiniLM	+0.24	[−0.10, +0.73]
ESCI new	BGE	−0.32	[−1.37, +0.58]

−4.94 [−6.96, −3.04] relative to hybrid. On historical ESCI, the selected strategy changes Recall@100 by +0.08 [−0.04, +0.19] percentage points relative to Original and −0.45 [−1.28, +0.39] relative to hybrid. Table 2 additionally tests the unchanged choice with MiniLM and newly sampled ESCI queries. These cells test transfer without encoder-specific retuning; historical test inspection and the new common-catalog construction limit confirmatory claims.

The WANDS MiniLM transfer changes recall by -1.75 $[-3.00, -0.63]$ points. Thus the development gain does not establish a transferable improvement; a simple fixed rule can remove incoming-order dependence and still reduce relevant-product inclusion.

The selected field rule is evaluated separately with one target changed at a time. On WANDS, its target-only macro Recall@100 change is -0.10 points (CI $[-1.37, +1.14]$), with 6.21% of highest-label pairs crossing this boundary. On ESCI, its target-only macro Recall@100 change is $+0.12$ points (CI $[+0.01, +0.24]$), with 0.45% of highest-label pairs crossing this boundary. A direct target gain does not imply the same catalog-wide gain, since competitors also change under platform deployment. No future query or test label is used to construct the rule; relevance labels only define the diagnostic population. The rule changes the raw-sort order of 38,934/42,994 WANDS products but only 527/10,076 historical ESCI products. ESCI’s largely unkeyed atoms make this literal transfer close to raw sorting; it is not evidence that semantic field priorities transfer across schemas.

Order invariance is not a recall guarantee. Set mean and canonical constructions remove input-order dependence, but their matching behavior differs. In the repaired historical comparison, set mean changes WANDS cNDCG@10 by -0.0096 (CI $[-0.0194, 0.0003]$) and ESCI by -0.0038 ($[-0.0147, 0.0071]$), using the corrected ESCI gains. These are small, statistically inconclusive effectiveness differences, not evidence of equivalence. Crucially, the new common-query ESCI comparison changes set-mean Recall@100 by -2.03 $[-3.15, -0.90]$ percentage points: the cNDCG result must not be mistaken for preserved recall. Direct set-mean-canonical cNDCG contrasts are $+0.0121$ ($[-0.0015, 0.0259]$) on WANDS and -0.0096 ($[-0.0210, 0.0020]$) on ESCI. Separate significance against Original would not establish superiority between these methods.

The fixed saturation check is more nuanced than a uniform failure: WANDS centroid recall is 0.6439 at four views and 0.6416 at seven, so adding views is not monotonically beneficial. The four-view change from Original is $+1.22$ points (CI $[+0.34, +2.29]$). These follow-up cells retain their uncertainty and cost; they do not reopen selection or establish cross-encoder generalization for a newly chosen winner.

All three historical late-aggregation variants remain order invariant yet reduce WANDS effectiveness, with paired intervals below zero. Their corrected ESCI intervals include zero. The negative variants and the fixed $m = 7$ saturation results remain in the appendix; extra views cannot retroactively change the selected strategy. Invariance supplies a design property, while recall and relevance determine whether that property is useful.

6.4 RQ4: What Are the Candidate and Computational Costs?

On WANDS, selected-method Recall@100 is 0.631; Original first meets or exceeds that value at $K = 100$ on the tested grid. On ESCI, selected-method Recall@100 is 0.911; Original first meets or exceeds that value at $K = 200$ on the tested grid. This is a discrete recall match, not an exact cost match. Rescued and newly missed pair counts are reported in Table 5 because similar net recall can conceal replacements of individual relevant products.

Table 3: Local exact retrieval on four CPU threads. Array storage is dense vectors plus sparse postings for hybrid, excluding text/vocabulary metadata. Time includes scoring, fusion where applicable, and stable sorting; query encoding is reported separately. These are not commercial latency estimates.

Data	Method	MiB	ms p50	ms p95
WANDS	Original	126.0	8.6	9.7
WANDS	BM25 + dense	176.2	15.2	18.0
WANDS	Appearance-first	126.0	8.6	9.8
ESCI	Original	29.5	2.1	2.6
ESCI	BM25 + dense	37.7	4.1	5.1
ESCI	Appearance-first	29.5	2.0	2.6

Table 4: Full-record view-count checks: held-out WANDS and historical ESCI Recall@100 and WANDS index/scoring cost. Centroids store one vector after multiple offline encodings; max stores all views. These fixed checks do not reopen strategy selection.

Method	R100 W	R100 E	MiB W	ms W
Centroid (2)	0.640	0.911	126.0	8.3
Max views (2)	0.640	0.911	251.9	13.1
Centroid (4)	0.644	0.910	126.0	8.3
Max views (4)	0.639	0.911	503.8	20.9
Centroid (7)	0.642	0.910	126.0	8.4
Max views (7)	0.642	0.910	881.7	36.7

Table 5: BGE Top-100 pair transitions relative to Original. Raw pair counts and query-macro net percentage points have different weighting.

Data	Method	Rescued	Missed	Net pp
WANDS	Hybrid	1485	1044	+4.89
WANDS	Appearance-first	669	640	-0.05
ESCI	Hybrid	108	59	+0.53
ESCI	Appearance-first	14	8	+0.08

The timing comparison separates index arrays, query encoding, and exact scoring/sorting. A centroid retains one vector per product after additional offline encoding; max aggregation retains all view vectors and repeats scoring. Every method returns K unique product IDs. Duplicate views are measured, not silently counted as independent evidence. Exact full-matrix scoring is largely insensitive to the requested K ; downstream reranking cost increases with the number of candidates. Thus equal K , equal index memory, and equal end-to-end cost are different comparisons.

Fixed-text reranking. The development recall gain from 200 to 500 candidates passes the frozen threshold; measured throughput also permits a Top-500 extension. Each method supplies the same fixed canonical product text to the same cross-encoder. The main table reports final highest-label Recall@20; returned-list condensed

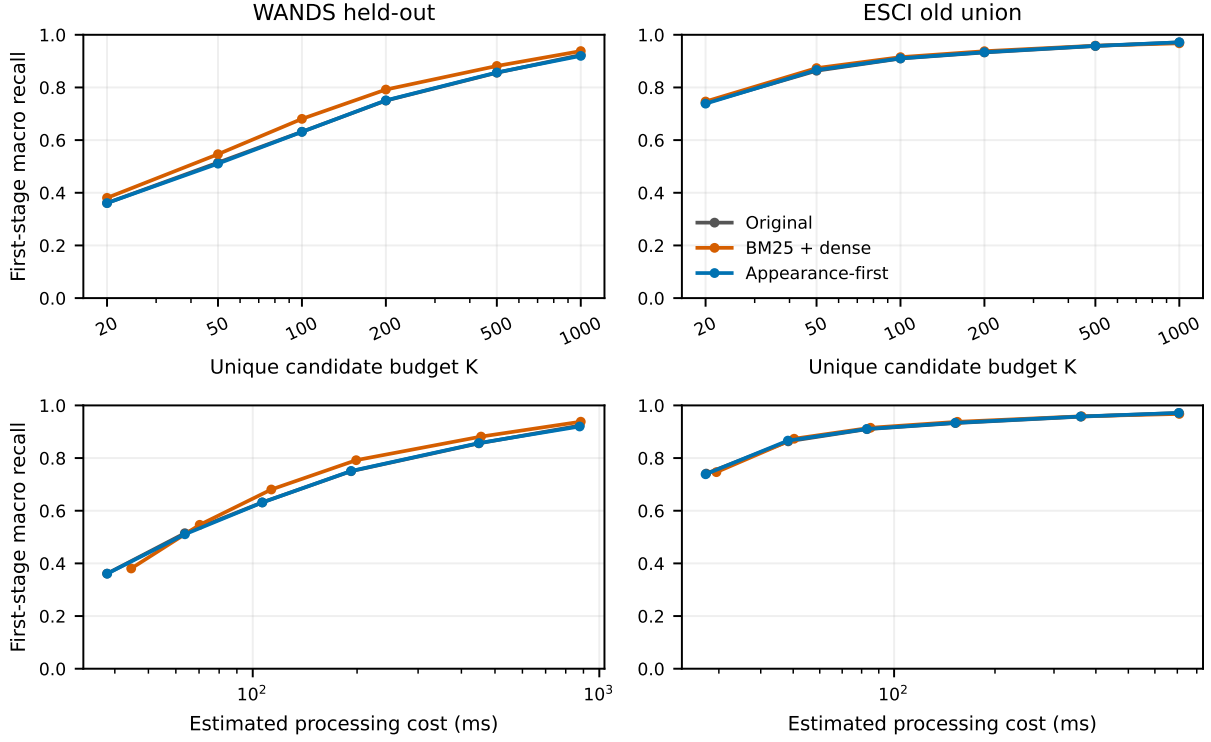


Figure 4: First-stage recall versus unique candidate budget (top) and estimated local processing cost (bottom). Cost adds median query encoding and exact retrieval to K times amortized cross-encoder pair time; it is a batch-throughput estimate, not measured online latency or final reranked recall. All curves reuse the same ranks.

Table 6: Highest-label Recall@20 after reranking a fixed canonical product text. Each column changes only the first-stage unique candidate budget. Same cross-encoder and final Top-20 across methods.

Data	First stage	K=50	100	200
WANDS	Original	0.376	0.368	0.377
WANDS	BM25 + dense	0.383	0.383	0.382
WANDS	Appearance-first	0.372	0.374	0.380
ESCI	Original	0.769	0.770	0.770
ESCI	BM25 + dense	0.771	0.773	0.775
ESCI	Appearance-first	0.770	0.770	0.770

NDCG and paired intervals are retained in the appendix. Amortized batched inference timing excludes model loading and is not an online request latency claim. Unlike first-stage Recall@K, final Top-20 recall need not grow monotonically with the candidate budget: additional candidates also compete for the fixed final slots. The WANDS point estimates illustrate this distinction; a dip between individual budgets is not automatically a significant loss. For WANDS at $K = 100$, the selected first stage changes final Recall@20 by +0.66 [-0.03, +1.58] points relative to Original. Increasing Original’s budget from 100 to 500 changes final recall by +1.13 [+0.08, +2.41] points. These paired contrasts separate the representation decision

from access to more candidates. For ESCI at $K = 100$, the selected first stage changes final Recall@20 by +0.02 [-0.10, +0.14] points relative to Original. Increasing Original’s budget from 100 to 500 changes final recall by +0.25 [-0.53, +1.03] points. These paired contrasts separate the representation decision from access to more candidates.

Reranking can correct ordering among candidates, but it cannot recover content it never receives. The legacy varying-input experiment reduced common-pool VI@20 from 25.94% to 13.03% on WANDS and from 3.95% to 1.37% on ESCI. That condition both varies candidate generation and exposes the reranker to different text; it is kept separate from the new canonical-input comparison. Its VI denominator includes only products surviving every pool, so it cannot be compared directly with all-highest VI in Figure 2. Reporting persistent omissions, candidate recall, and final recall prevents conditional reranker stability from being mistaken for recovery.

7 Discussion and Limitations

7.1 Robustness, Recall, and Deployment Scope

Representation-induced instability is a failure mode, not a complete optimization objective. Stabilizing an index can preserve persistent omission; maximizing over more views can raise irrelevant scores as well as relevant ones. A credible improvement must therefore be

judged against simple canonical and hybrid baselines, unchanged strategy transfer, candidate budgets, and computational costs. Our experiments separate merchant-editable text from platform-only aggregation. Single-target effects do not establish a collective benefit when every competitor adopts the same rule.

These results suggest a concrete evaluation practice for structured Web indexing. First verify that an intervention preserves complete source facts, then report both cutoff crossings and persistently missing relevant items. Evaluate a local record replacement separately from changing the entire catalog, and retain query-level gains and losses rather than only their average. Finally, compare candidate and final-stage outcomes at declared resource budgets. A deterministic serializer supplies a reproducible interface, but choosing one ordering does not establish that it retrieves the right content. This distinction matters when a downstream agent only receives a small candidate set and cannot inspect the omitted catalog.

7.2 Scope of the Visibility Claim

Visibility means eligibility at an offline retrieval cutoff. It is not observed exposure, clicks, satisfaction, purchase, or welfare, and human relevance labels are not objective product quality. The shopping-agent motivation explains why candidate inclusion matters but does not replace an agent evaluation. The mechanism may arise wherever unordered records are serialized for neural retrieval, including jobs, hotels, restaurants, and events; neither its existence nor its magnitude in those domains is established here.

7.3 Limitations and Evaluation History

The evidence covers two English public datasets, synthetic fact-preserving permutations, fixed catalogs, and offline exact retrieval. ESCI judgments are incomplete and the historical union differs from its official small candidate pools. New query IDs were frozen before evaluation, but the source dataset and historical test results had already been accessed. Query holdout is not unseen-product or unseen-category generalization. Empty query prefixes retain the historical model condition; optimized prompts, retraining, ANN indexing, commercial search, and live users remain untested. Rank strata have limited common support and residual margin confounding; no matched attribute-position experiment supports an internal attention mechanism. No non-inferiority margin or broad method-superiority claim follows from intervals that include zero.

8 Conclusion

Fact-equivalent representations can change which judged-relevant products enter a retrieval pipeline, even when only one product changes and every competitor remains fixed. Persistent omission shows why eliminating order instability is insufficient. The development-selected ordering rule does not deliver a consistent recall gain and loses recall on WANDS when transferred to another encoder; simple hybrid retrieval remains a strong comparator. Fixed text rules and platform aggregation must therefore be evaluated against candidate-budget and cost controls, not credited for invariance alone. The resulting principle is to measure representation robustness together with relevant-product recall and the resources required to obtain it.

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A Audit, Selection, and Additional Results

The frozen protocol and explicit query IDs precede this extension’s strategy evaluation, but do not preregister the historical study. The archived prior manuscript and result artifacts remain unchanged. Gain reconciliation updates effectiveness from fixed ranks; it does not silently replace the old ESCI gain convention. The new exact rankings use stable catalog-index ties. Legacy candidate reconciliation finds small numerical/order differences but no changed highest-label Top-100 membership in any original serialization condition.

Negative results and strategy boundaries. Table 7 retains every screening outcome. Table 9 reports both aggregation families at every fixed view count without reopening selection. Canonical-anchored view generation is a deterministic input-order-independent construction, not proof that arbitrary sampled averaging is invariant. All methods preserve complete raw atoms;

Table 7: All eight added configurations on the frozen WANDS development queries. Sorting reflects the prespecified selection rule, not test performance.

Strategy	R100	cN@10	Vectors
Appearance-first	0.6800	0.8353	1
max 4	0.6735	0.8484	4
centroid 4	0.6712	0.8442	1
Dimension-first	0.6693	0.8387	1
max 2	0.6679	0.8485	2
centroid 2	0.6670	0.8465	1
Type-first	0.6605	0.8388	1
Use-first	0.6597	0.8369	1

text and encoded-vector multiplicities differ when small attribute sets or encoder truncation produce duplicate views. Set mean encodes atoms independently and therefore differs from full-record centroids and max retrieval.

Meaning of candidate coverage. Table 11 explicitly includes persistently absent pairs. At Top-100 the historical common-pool reranker analysis excludes those pairs as well as sometimes-retrieved pairs. Its apparent stability must be interpreted conditional on survival. Table 5 separately exposes recovered and newly lost pairs; the full artifact provides these quantities per query at every budget, normalized by the same highest-label population.

B Reranking and Reproduction

The first-stage order arrays are reduced to unique product IDs before candidate selection. All candidate sources use identical canonical product text, model weights, truncation, and tie handling for reranking. The larger-budget decision is based on development recall and measured compute, not on a favorable held-out final-rank result. Full per-query metrics, final Top-20 IDs, pairwise cross-encoder scores, and paired bootstrap contrasts accompany Table 12. Candidate-union caching reuses a pair’s inference across budgets and methods.

Model cache keys include input-content hashes. The reproduction scripts retain split IDs, field-priority keywords, product-dependent view seeds, encoder revisions, native pooling and normalization, official-gain repair, exact ranking, and local timing records. Nine integrity tests cover existing metric reconciliation, target insertion against explicit reranking, preserved atom multiplicity, independent metric eligibility, normalization, and product-level max reduction. Timings distinguish offline encoding, query encoding, score/sort operations, and amortized reranker inference. No ANN approximation is included.

Historical set mean encoded queries in a separate cached pass. The extension instead shares the frozen Original query matrix across all methods; the audit attributes tiny baseline score differences to these query embeddings. Historical repaired comparisons retain their original ranks, while new tables use the common query matrix and stable ranking. These two conditions are explicitly distinguished rather than silently overwriting a validated result.

Table 8: Primary seven-order VI@20 percentages. Catalog-wide and target-only interventions are distinct. Micro weights pairs; macro weights eligible queries. Query-bootstrap intervals remain in the corresponding source tables and main visualization.

Data	Encoder	Catalog micro	Catalog macro	Target micro	Target macro
WANDS	MiniLM	13.26	19.17	12.72	18.96
WANDS	BGE	9.41	12.05	8.84	11.36
WANDS	GTE	11.33	14.43	11.39	14.60
ESCI	MiniLM	4.67	4.04	3.11	2.91
ESCI	BGE	4.42	3.66	3.16	2.42
ESCI	GTE	8.25	6.91	6.68	5.88

Table 9: Fixed BGE view-count checks on held-out WANDS and historical ESCI. Differences are Recall@100 percentage points from Original. These checks do not change the selected strategy.

Data	Method	R100	Δ	95% CI
WANDS	centroid 2	0.640	+0.78	[-0.11,+1.79]
WANDS	max 2	0.640	+0.83	[+0.02,+1.80]
WANDS	centroid 4	0.644	+1.22	[+0.34,+2.29]
WANDS	max 4	0.639	+0.69	[-0.30,+1.80]
WANDS	centroid 7	0.642	+0.99	[+0.02,+2.07]
WANDS	max 7	0.642	+0.98	[-0.11,+2.23]
ESCI	centroid 2	0.911	+0.09	[-0.08,+0.25]
ESCI	max 2	0.911	+0.11	[-0.07,+0.28]
ESCI	centroid 4	0.910	+0.04	[-0.12,+0.19]
ESCI	max 4	0.911	+0.07	[-0.11,+0.25]
ESCI	centroid 7	0.910	+0.04	[-0.12,+0.18]
ESCI	max 7	0.910	+0.07	[-0.11,+0.25]

Table 10: Reconciled historical mitigation cNDCG@10 and paired differences from Original. ESCI uses E/S/C/I=1/.1/.01/0. WANDS uses 383 positive-IDCG held-out queries; ESCI uses 500.

Data	Method	cN@10	Δ	95% CI
WANDS	Original	0.8247	+0.0000	[+0.0000,+0.0000]
WANDS	Canonical	0.8030	-0.0217	[-0.0333,-0.0105]
WANDS	Factual	0.8178	-0.0069	[-0.0139,-0.0000]
WANDS	Set mean	0.8151	-0.0096	[-0.0194,+0.0003]
WANDS	Late mean	0.8058	-0.0189	[-0.0293,-0.0083]
WANDS	Late max	0.8093	-0.0153	[-0.0266,-0.0045]
WANDS	Late top3	0.8106	-0.0141	[-0.0243,-0.0038]
ESCI	Original	0.7364	+0.0000	[+0.0000,+0.0000]
ESCI	Canonical	0.7423	+0.0058	[-0.0012,+0.0131]
ESCI	Factual	0.7305	-0.0059	[-0.0127,+0.0007]
ESCI	Set mean	0.7326	-0.0038	[-0.0147,+0.0071]
ESCI	Late mean	0.7290	-0.0074	[-0.0181,+0.0031]
ESCI	Late max	0.7297	-0.0067	[-0.0180,+0.0046]
ESCI	Late top3	0.7290	-0.0074	[-0.0181,+0.0031]

C Illustrative Case and Diagnostic Scope

The motivating query is *turquoise chair* (WANDS query 162); the Exact-relevant item is product 34536, titled “adjustable swivel office chair computer chair task chair mesh chair , turquoise.” Its raw attributes include “framecolor : green,” “weightcapacity:250,” and

“framematerial : steel.” The apparent title/attribute color discrepancy is present in the source; we do not repair or rewrite it. The full atom list and all unchanged free text are stored in the case artifact. This is an intentionally inspectable, post-hoc illustration, not a representative effect-size estimate.

The fitting-input diagnostic requires every tested serialization to fit the encoder budget. Coarse rank bands compare relevance strata only where both occur in the same query. Additional descriptive bins cover score margin, maximum input length, attribute multiplicity, query length, and literal value overlap. These bins characterize association and support; they do not establish an internal positional mechanism. The earlier official ESCI judged-pool check is retained in the artifact, but its median pool of 16 cannot substantiate large-catalog Recall@100.

Table 11: BGE seven-order coverage. Counts are highest-label query–product pairs; macro percentages average within eligible queries. All historical queries are used here, unlike strategy selection/held-out tables. Sometimes means crossing; never is persistent omission.

Data	K	Always	Some	Never	Always %	Some %	Never %
WANDS	20	2158	2396	20916	30.62	12.05	57.33
WANDS	50	4997	3414	17059	45.28	12.77	41.95
WANDS	100	8384	3856	13230	57.58	12.52	29.90
WANDS	200	12683	3604	9183	69.99	9.74	20.27
WANDS	500	17661	3246	4563	82.39	7.49	10.12
WANDS	1000	21562	1992	1916	90.15	4.41	5.43
ESCI	20	2907	196	1331	71.89	3.66	24.44
ESCI	50	3751	94	589	85.54	2.20	12.26
ESCI	100	3995	44	395	90.73	0.72	8.55
ESCI	200	4125	38	271	93.00	0.81	6.19
ESCI	500	4245	20	169	95.33	0.53	4.14
ESCI	1000	4306	16	112	96.92	0.37	2.71

Table 12: Fixed canonical-input reranking. Each K cell gives Recall@20 / returned-list condensed NDCG@10; the latter is not full-catalog cNDCG. Multiply measured amortized ms/pair by K to estimate candidate inference time, excluding loading and retrieval. This is batch throughput, not per-request latency. Paired intervals are in the artifact.

Data	Method	K=50	K=100	K=200	K=500	ms/pair
WANDS	Original	0.376 / 0.786	0.368 / 0.784	0.377 / 0.777	0.379 / 0.776	0.857
WANDS	Hybrid	0.383 / 0.787	0.383 / 0.785	0.382 / 0.779	0.385 / 0.777	0.857
WANDS	Appearance-first	0.372 / 0.784	0.374 / 0.785	0.380 / 0.783	0.384 / 0.779	0.857
ESCI	Original	0.769 / 0.692	0.770 / 0.692	0.770 / 0.693	0.773 / 0.694	0.694
ESCI	Hybrid	0.771 / 0.694	0.773 / 0.695	0.775 / 0.694	0.774 / 0.694	0.694
ESCI	Appearance-first	0.770 / 0.693	0.770 / 0.693	0.770 / 0.693	0.772 / 0.694	0.694

Table 13: Actual unique views per product. Exact vector equality can merge different complete texts after truncation; it is not a tolerance-based semantic equivalence test.

Data	m	Text mean	Vec. mean	One vec.
WANDS	2	2.00	2.00	66
WANDS	4	4.00	4.00	59
WANDS	7	7.00	6.99	59
ESCI	2	1.67	1.62	3819
ESCI	4	2.59	2.47	1968
ESCI	7	3.40	3.21	1672

Table 14: All orders for the Exact-relevant turquoise chair (MiniLM). Target-only competitors stay in C0. Positive Δ means a worse rank than target C0.

Order	Target-only rank	Δ vs C0	Catalog-wide rank
C0	486	0	486
C1	476	−10	295
C2s1	1,527	+1,041	1,303
C2s2	28	−458	21
C2s3	582	+96	484
C2s4	5	−481	3
C2s5	5	−481	4